

Celebal Excellence Internship

[Week – 4 Task – Azure cloud concepts]

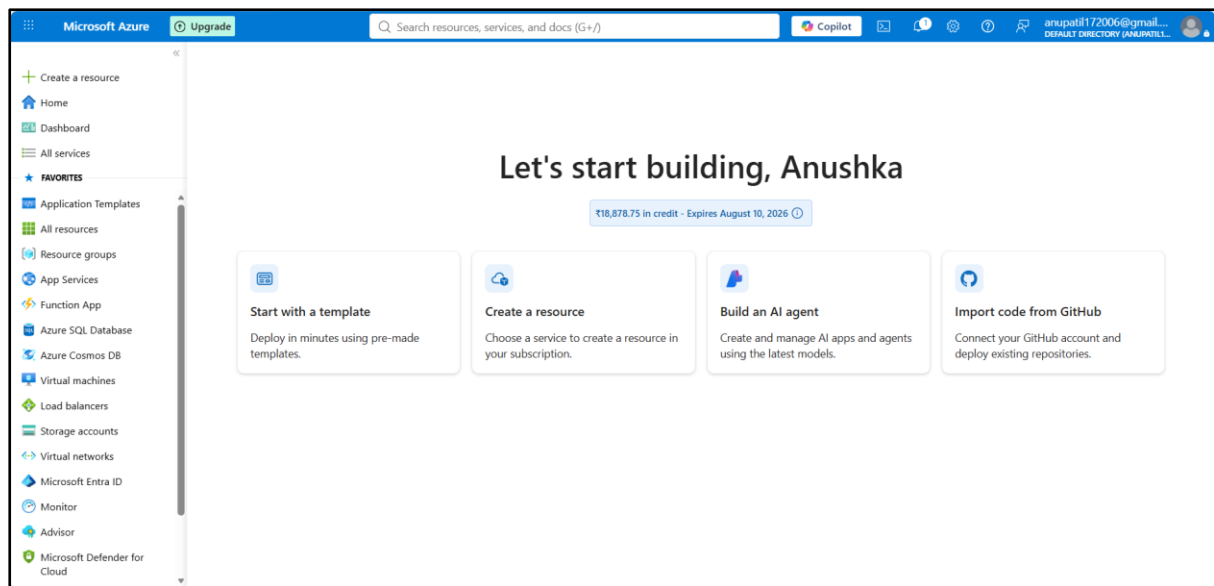
Title: CEI – week4 task submission

Name: Anushka Manoj Patil

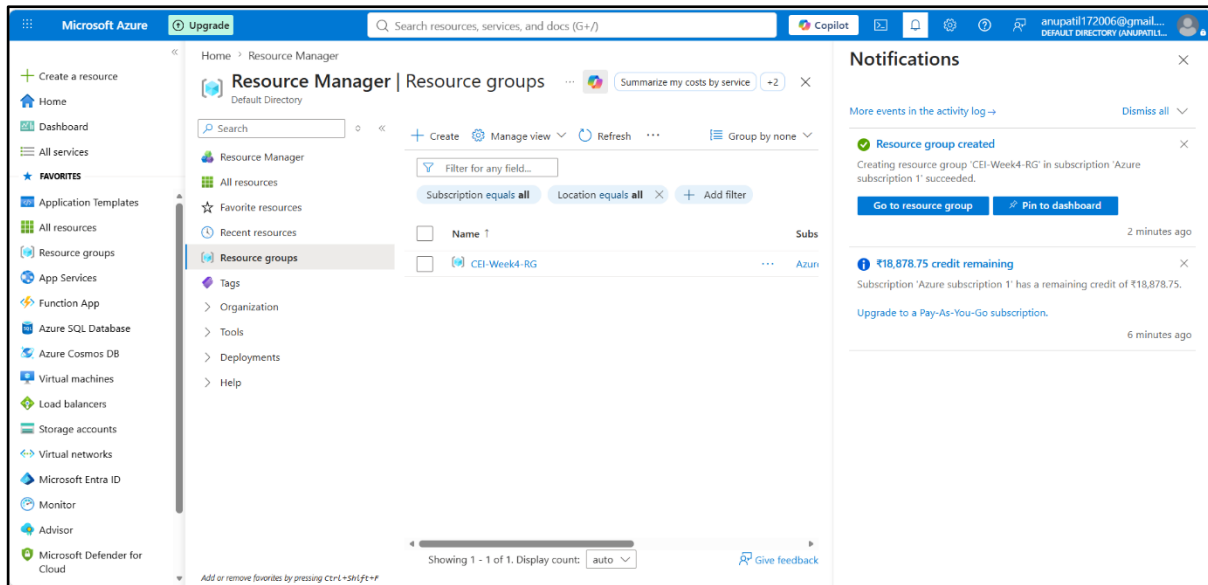
College name: Dr. D. Y. Patil Pimpri, Pune

Step 1: Explore Azure Portal and create a Resource Group.

1. 1 – Explore Azure Portal:



1.2 – Create a Resource Group

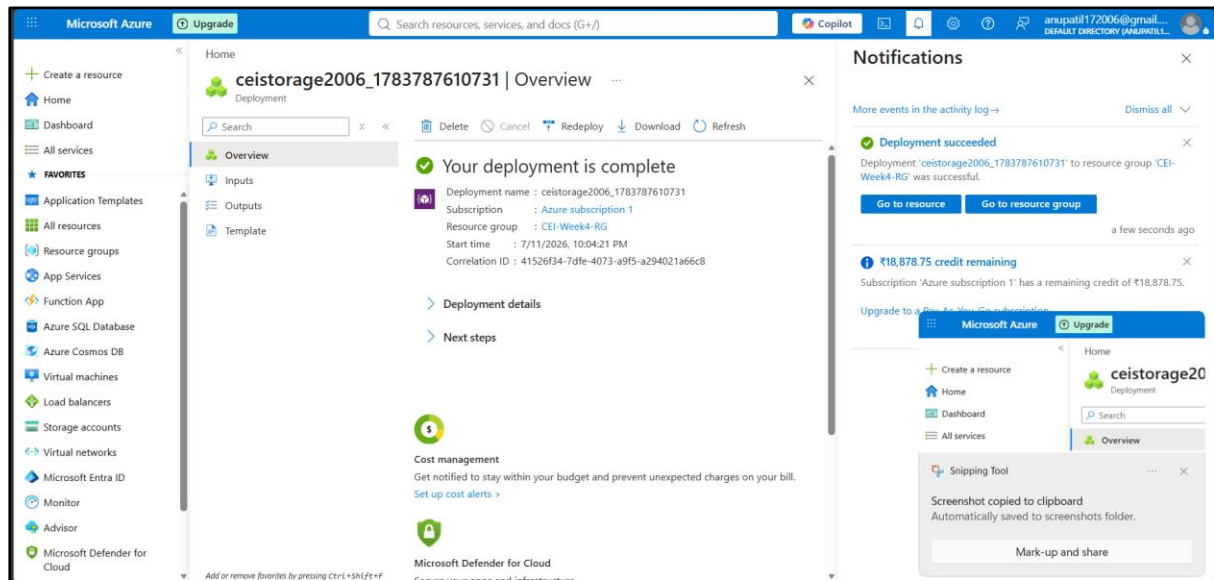


Insight:

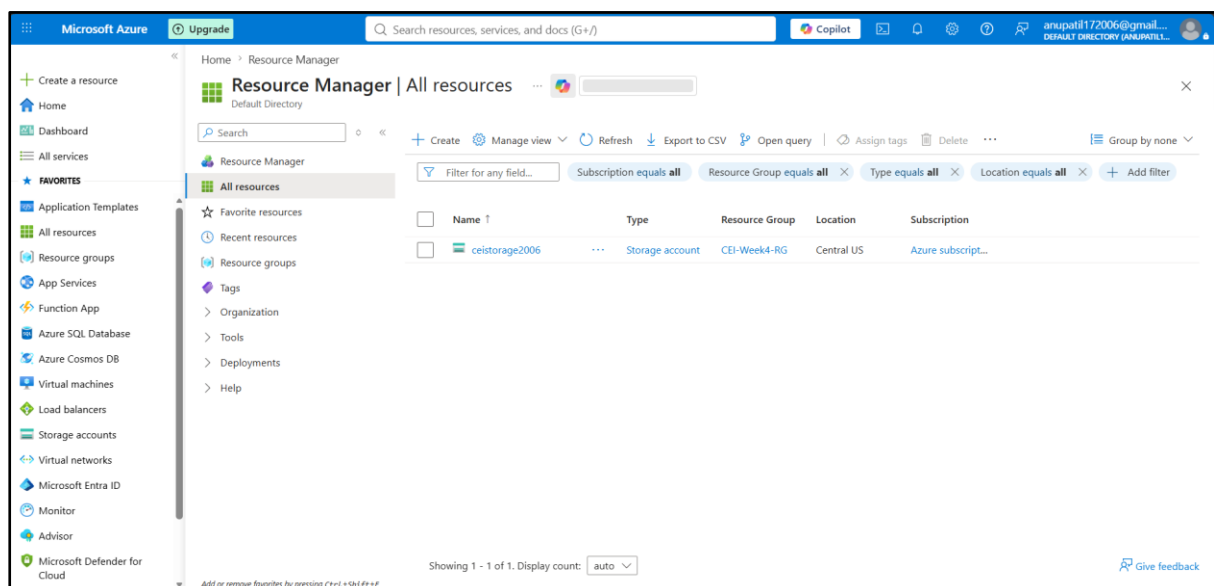
- Explored the Azure portal interface.
- Understood Azure's resource management hierarchy.
- Created a Resource Group to manage project resources efficiently.

Step 2: Storage Setup

2.1 – Create a Storage Account



2.2 – Open the Storage Account

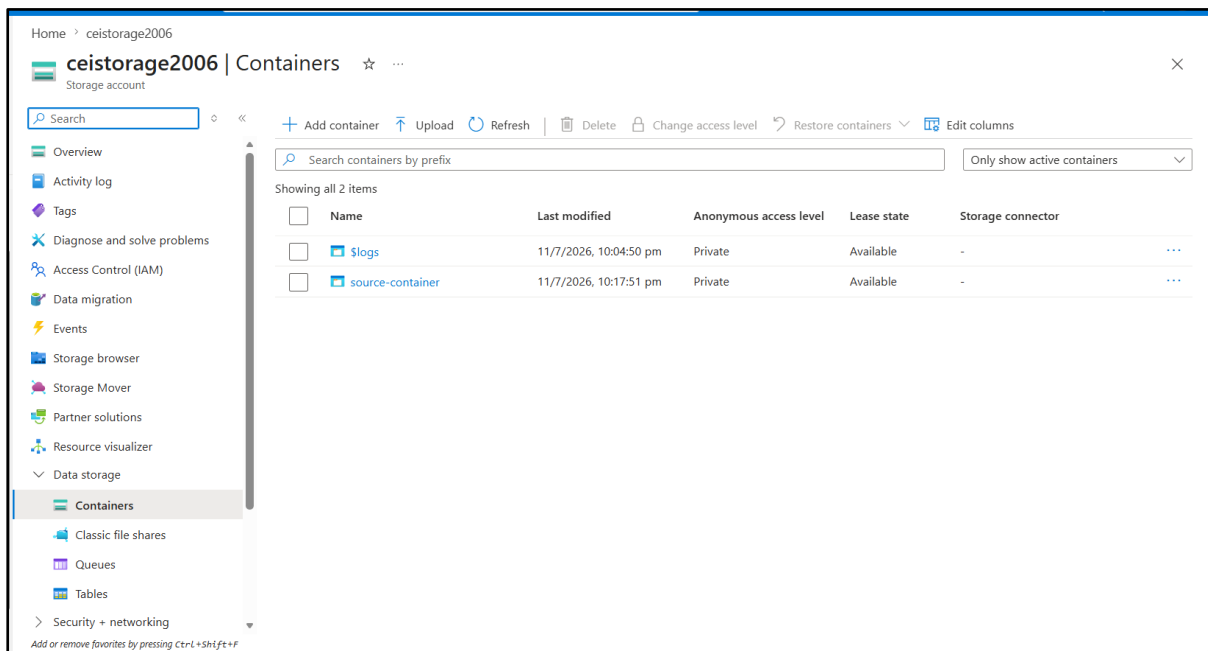


Insight:

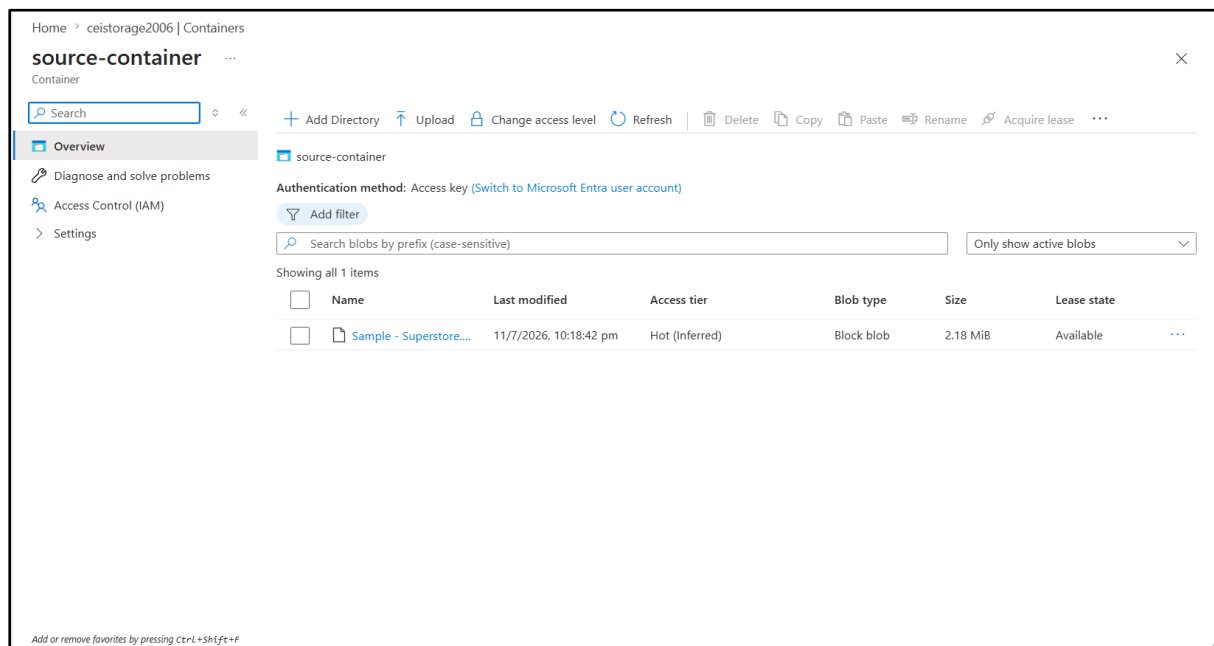
- Learned how Azure Storage provides scalable and secure cloud storage.
- Configured a Storage Account using the Standard performance tier and LRS redundancy.
- Understood that Blob Storage is commonly used for storing datasets used in Azure Data Factory pipelines.

Step 3: Create a Blob Container & Upload a CSV File

3.1 Creating blob container.



3.2 Uploading the CSV file.

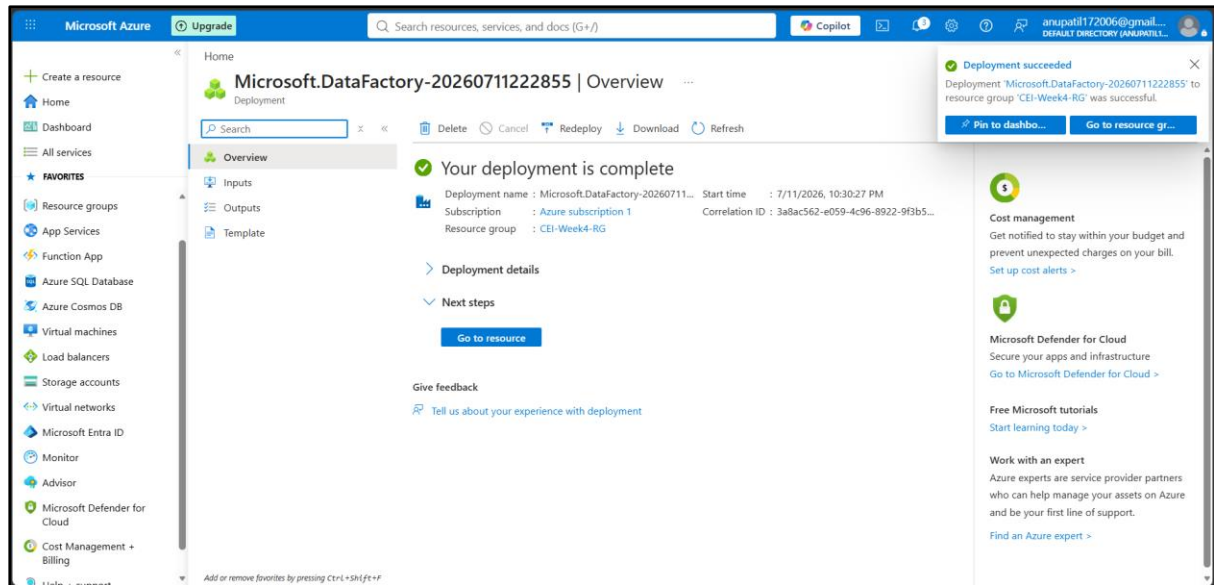


Insight:

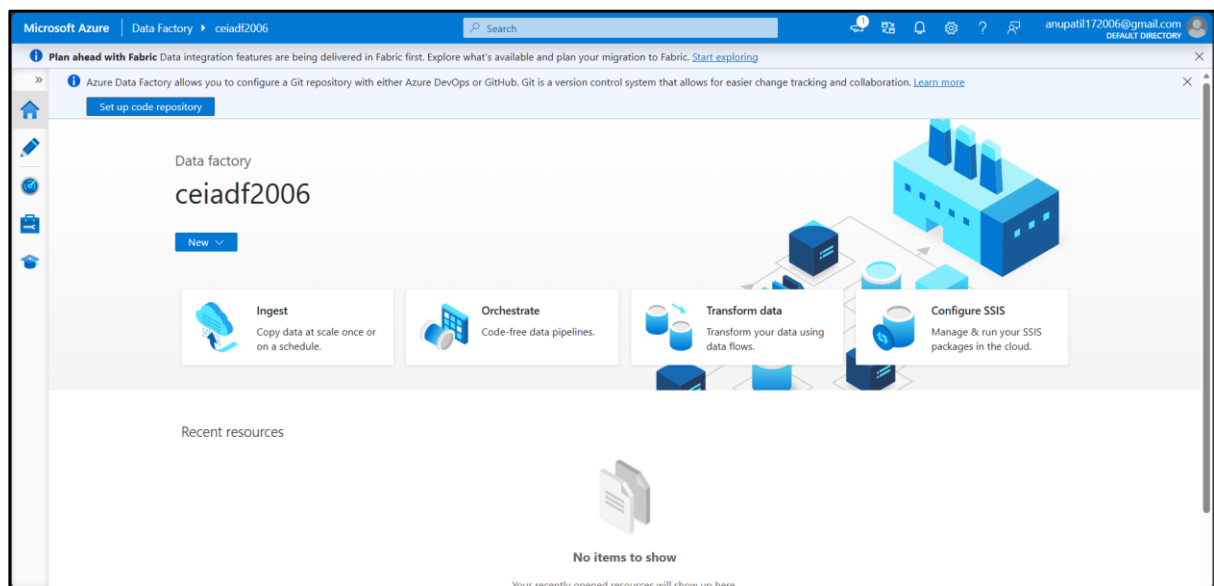
- Created a Blob Container (source-container) inside the Azure Storage Account.
- Uploaded the source CSV dataset to Azure Blob Storage. This container will serve as the source for the Azure Data Factory pipeline.

Step 4: Creating Azure Data Factory(ADF)

4.1 Creating Azure data factory



4.2 Opening data Factory



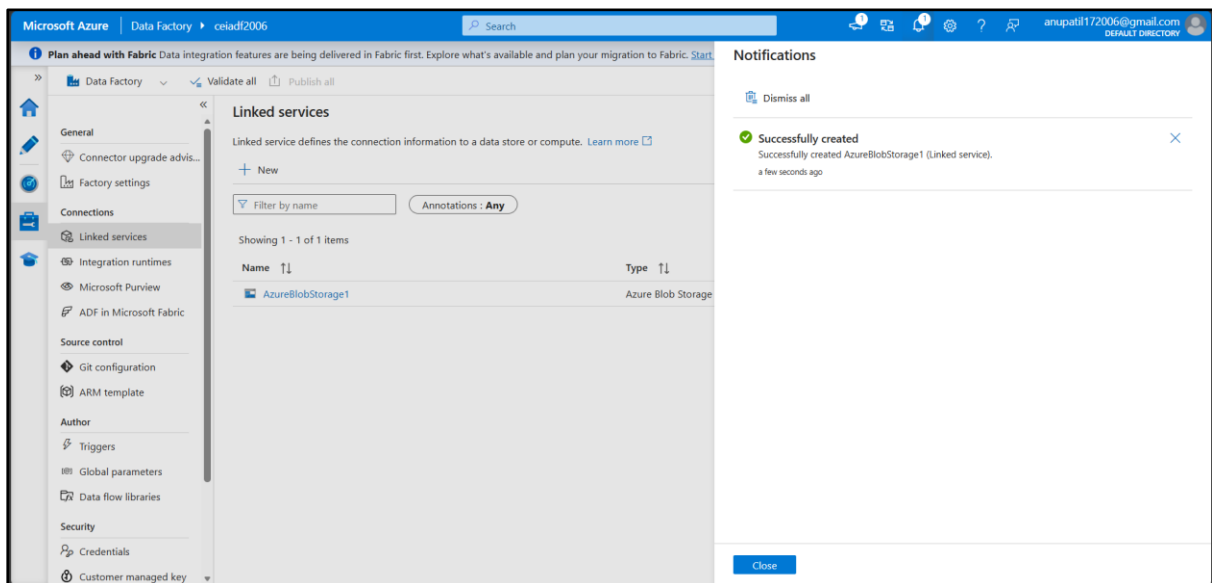
Insight:

- Created an Azure Data Factory (V2) instance within the same Resource Group and Region as the Storage Account.

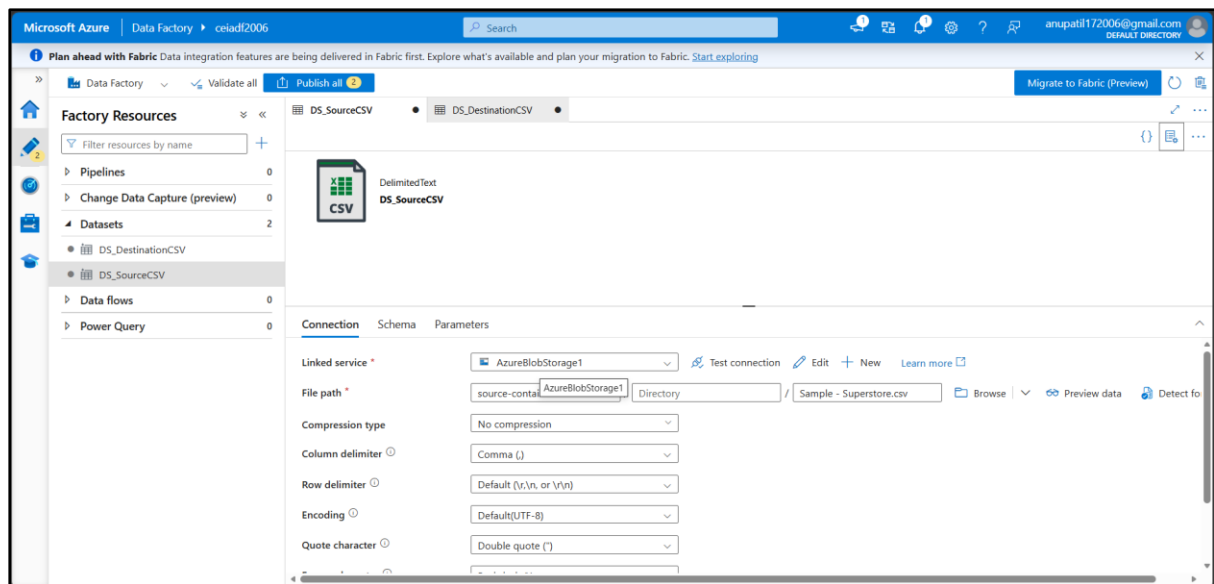
- Launched Azure Data Factory Studio and explored the **Author**, **Monitor**, and **Manage** modules, which are used for developing, monitoring, and administering data integration pipelines. Understood that Blob Storage is commonly used for storing datasets used in Azure Data Factory pipelines.

Step 5: Create Linked Service and Datasets

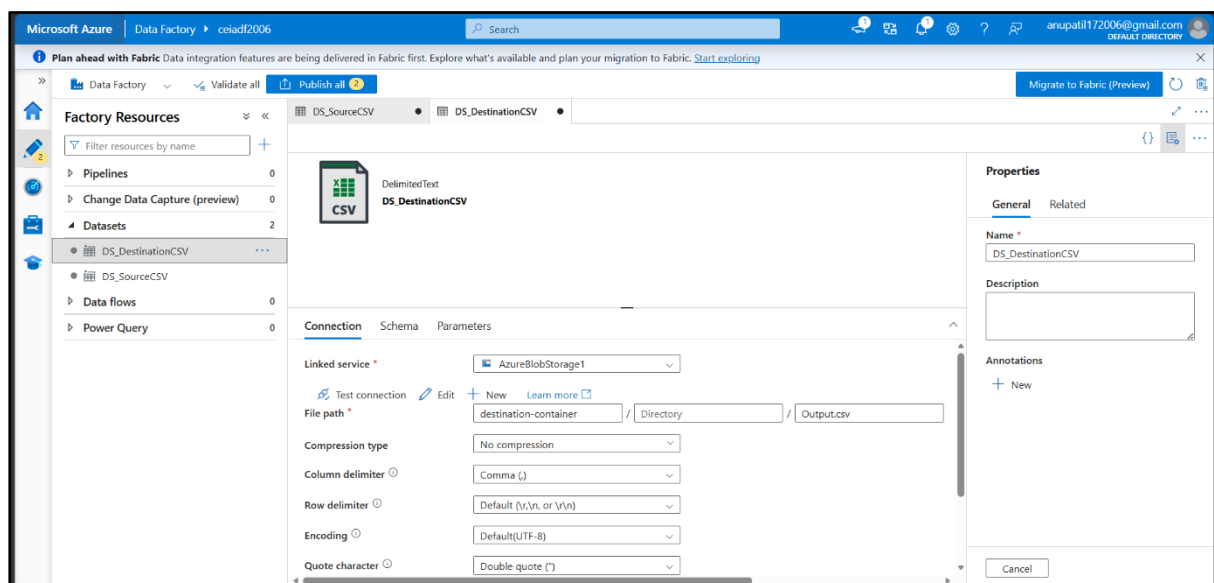
5.1 Creating linked service



5.2 Creating Source Dataset



5.3 Creating Destination Dataset

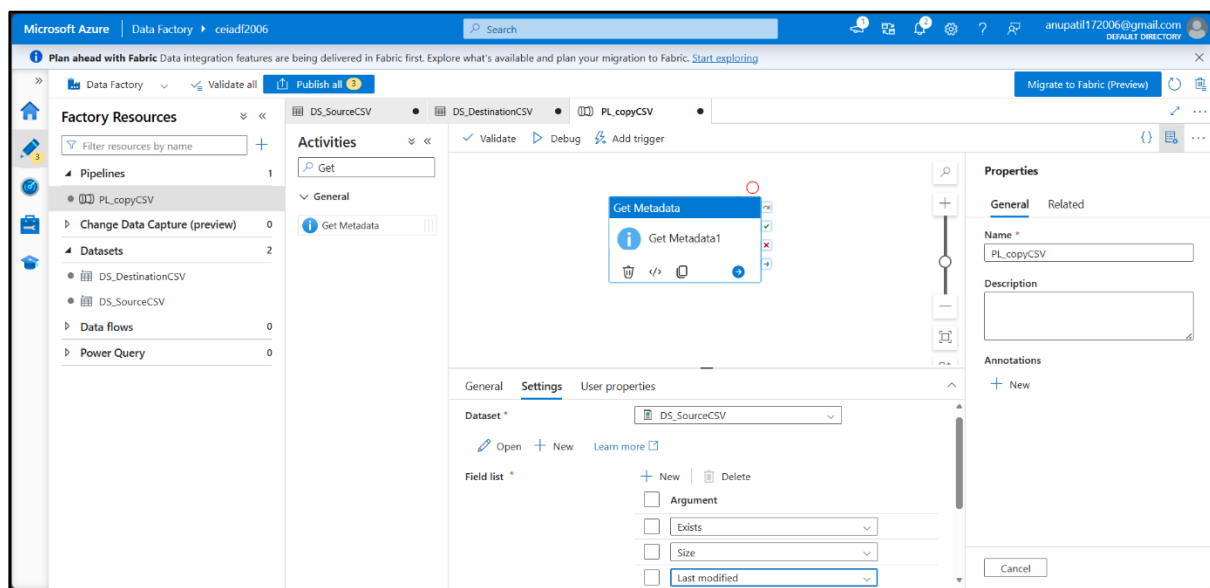


Insight:

- Created a **Linked Service** to establish a secure connection between Azure Data Factory and Azure Blob Storage.
- Configured a **Source Dataset** pointing to the uploaded CSV file.

- Configured a **Destination Dataset** to store the copied data generated by the pipeline.

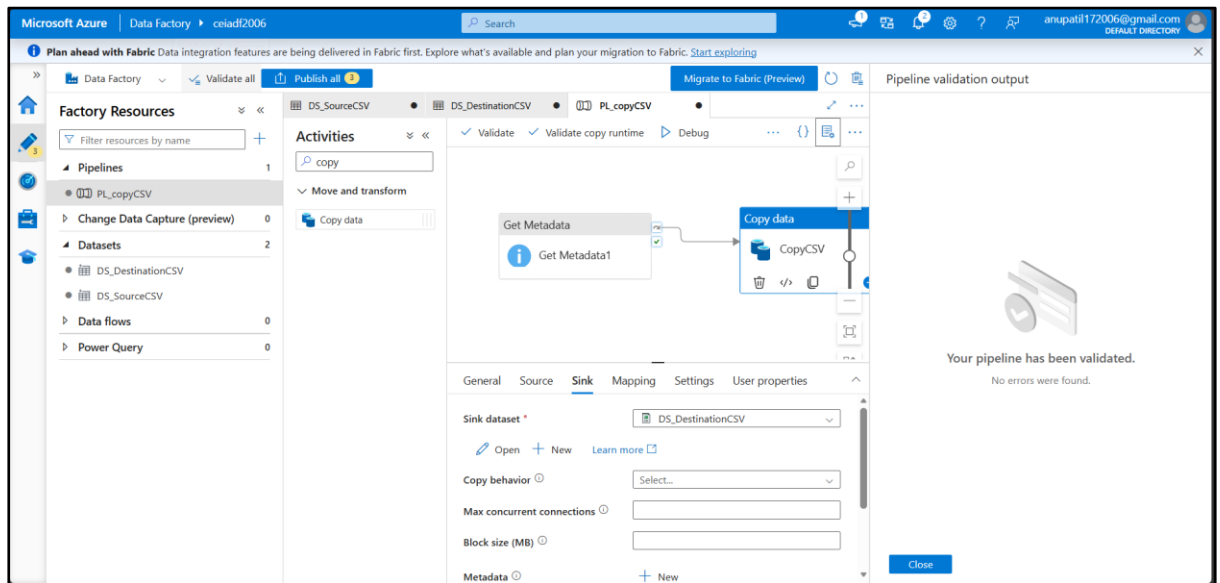
Step 6: Creating Pipeline and Add Get Metadata Activity



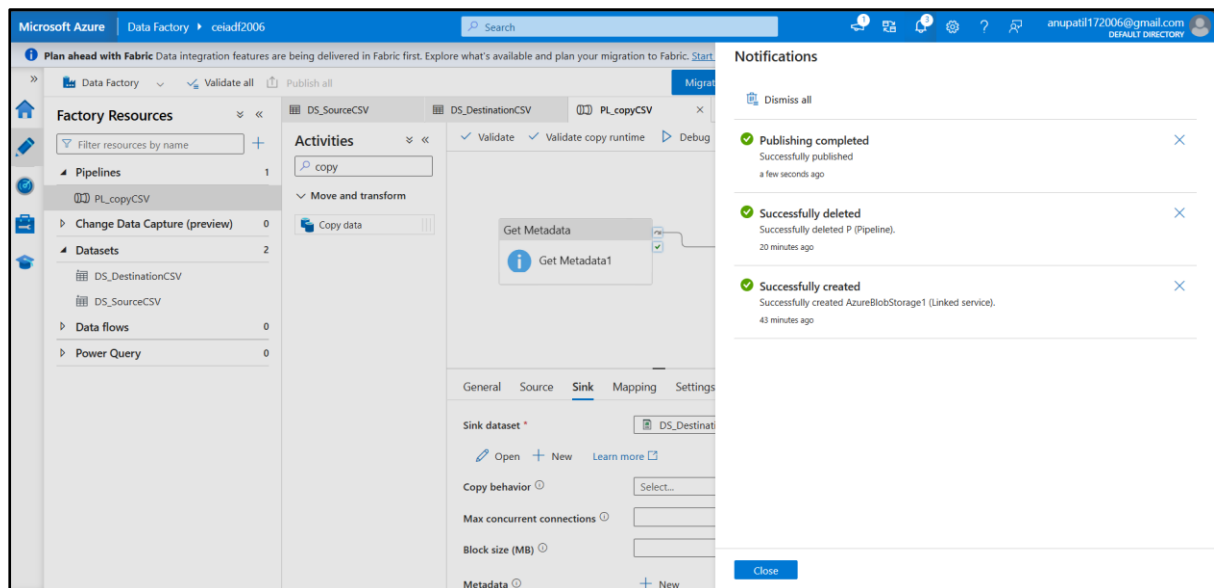
Insight:

Created a pipeline named **PL_CopyCSV** and added a **Get Metadata** activity to validate the source file before processing. The activity retrieves file metadata such as size and last modified time to ensure the file is available for copying.

Step 7: Get Metadata + Copy Data



7.1 Pipeline Validation



Insight:

Created an Azure Data Factory pipeline using **Get Metadata** and **Copy Data** activities. The pipeline first validates the source file by retrieving metadata and

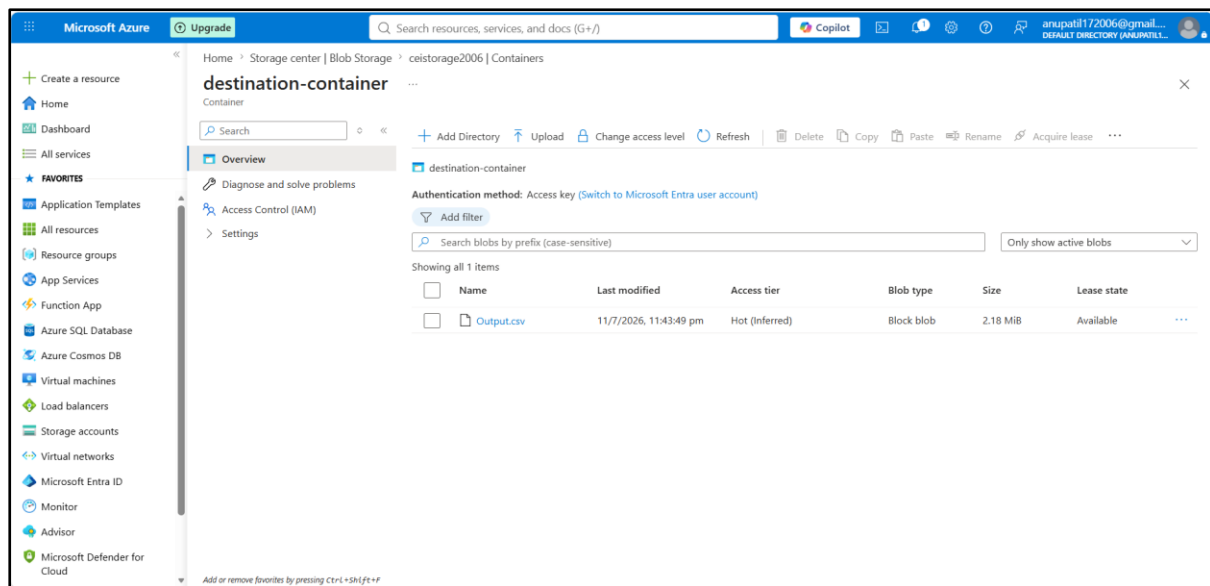
then copies the CSV file from the source Blob container to the destination Blob container.

Step 8: Run the pipeline

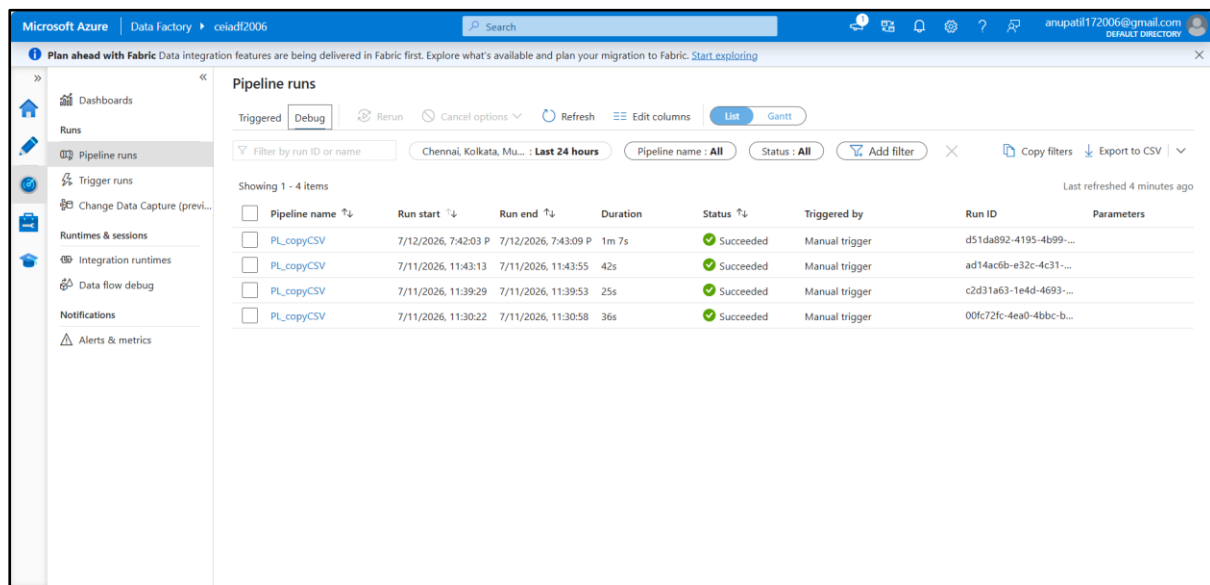
The screenshot displays the Microsoft Azure Data Factory console. On the left, the 'Factory Resources' pane shows a list of resources including Pipelines (1), Datasets (2), Data flows (0), and Power Query (0). The 'Pipelines' section is expanded, showing the pipeline 'PL_copyCSV'. The main canvas shows the pipeline design with two activities: 'Get Metadata1' and 'CopyCSV', connected by a dependency arrow. The 'Activities' pane on the right shows the 'Copy data' activity selected. Below the canvas, the 'Output' tab is active, displaying the pipeline run ID 'ad14ac6b-e32c-4c31-b442-59677c91c075' and a status of 'Succeeded'. A table below shows the execution details for the activities.

Activity name	Activity status	Activity name	Run start	Duration	Integration runtime
CopyCSV	Succeeded	Copy data	7/11/2026, 11:43:37 PM	16s	AutoResolveIntegrationRuntime (Centr
Get Metadata1	Succeeded	Get Metadata	7/11/2026, 11:43:19 PM	16s	AutoResolveIntegrationRuntime (Centr

8.1 Pipeline verification



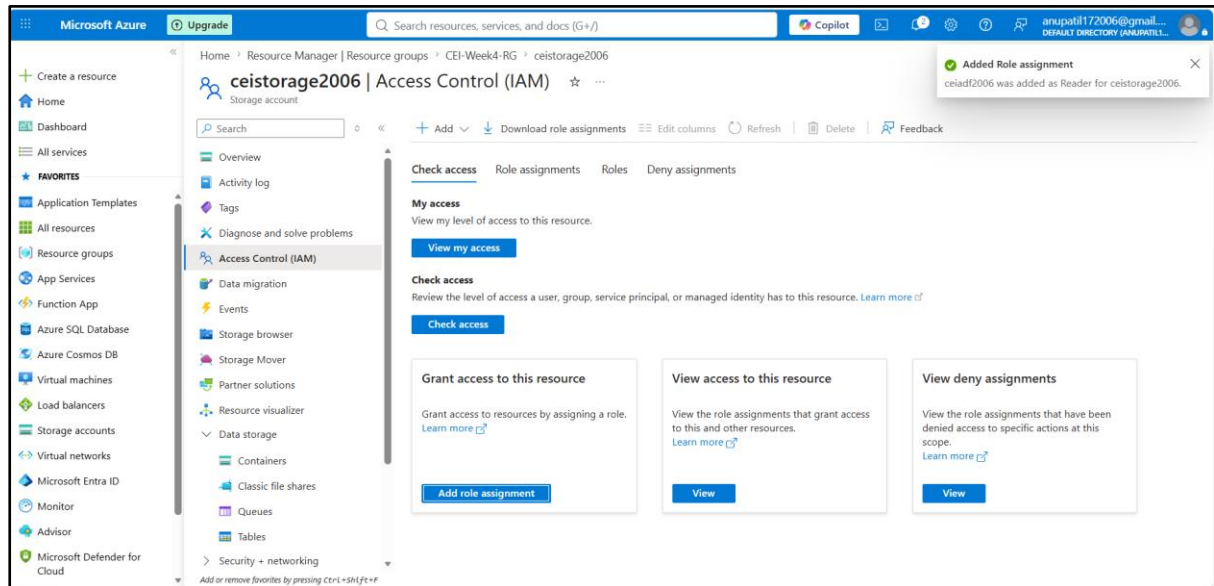
8.2 Verification success



Insight:

- Published the Azure Data Factory pipeline.
- Executed the pipeline using **Debug**.
- Verified successful execution of **Get Metadata** and **Copy Data** activities before monitoring the pipeline run.

Step 9: Assigning IAM role



Insight:

- Opened the Storage Account's **Access Control (IAM)**.
- Started assigning Azure RBAC roles to provide secure access between Azure Data Factory and Azure Storage.
- Used Azure's role-based access control (RBAC) to manage permissions.

Brief Summary:

This assignment involved creating an end-to-end data integration pipeline using Microsoft Azure services. A Resource Group, Storage Account, Blob Storage containers, and Azure Data Factory were successfully created. A CSV file was uploaded to the source container, and a pipeline consisting of Get Metadata and Copy Data activities was designed. The pipeline was validated, debugged, and executed successfully, copying the source CSV file to the destination container. Finally, the required IAM role was assigned, and the pipeline execution was verified through Azure Data Factory. This assignment provided practical experience in building and managing cloud-based data pipelines using Azure.